

Sensor stack (feeds the agent)



Incoming call | 8 kHz



Preprocessor | telephone-band channel | VAD



Acoustic stream

residual | prototypes | CUSUM



Linguistic stream

keywords | stage tracker
(no production ASR here)



Fusion | regimes | SAPC | ACI

sufficient statistics only (no waveform, no transcript)



Defense agent (closed loop, Fig. 2)